

9 Application and Implementation

备注

以下应用部分中的信息不属于 TI 器件规格的范围，TI 不担保其准确性和完整性。TI 的客户应负责确定器件是否适用于其应用。客户应验证并测试其设计，以确保系统功能。

9.1 Application Information

The BQ769x0 family of battery monitoring AFEs enabling cell parametric measurement and protection is a variety of 3-series to 15-series Li-ion/li-polymer battery packs.

To evaluate the performance and configurations of the device users need the , [BQ76940](#), [BQ76930](#), and [BQ76920 Evaluation Software](#) tool to configure the internal registers for a specific battery pack and application. The Evaluation Software tool is a graphical user-interface tool installed on a PC during development. This can be used in conjunction with the BQ76920EVM, BQ76930EVM or BQ76940EVM.

The BQ769x0 devices are expected to be implemented in a system with a microcontroller that can perform additional functions based on the data made collected. The [BQ78350-R1](#) is one example of a companion to the BQ769x0 family. Additional application information is available in the [BQ769x0 Family Top Design Considerations Application Report](#).

9.1.1 Device Timing

The device timeline accuracy is typically 3.5%. Each five-cell group in the BQ76930 and BQ76940 devices uses an independent 250-ms timeline, so voltage and temperature measurements of different groups drift with respect to one another.

9.1.2 Random Cell Connection

The device design anticipated transient conditions during cell connection, but that design did not result in unique specifications. The large component value ranges used in the application circuits may require special considerations for random cell connection. See additional information in the [BQ769x0 Family Top Design Considerations Application Report](#).

9.1.3 Power Pin Diodes

The VC5X, VC10X, and BAT pins must have a diode from the top group input to the associated power pin, as shown in [图 9-2](#) and [图 9-3](#). These diodes limit the excursion of the input voltage above the supply. The diodes should be conventional diodes rather than Schottky type to allow some variation of the supply voltage without loading the cell input. When needed, two diodes may be used in series.

9.1.4 Alert Pin

The ALERT pin is an input and output. The input is sensitive to noise and may benefit from a RC filter at the pin to reduce noise at the pin. A maximum 250- μ s time constant is suggested to allow the pin to fall when it is cleared as an output before it is sampled as an input. 500-k Ω and 470-pF values are commonly recommended. Guard traces around the traces may be helpful to avoid crosstalk to the ALERT trace.

9.1.5 Sense Inputs

The SRP input should operate near VSS, so VSS references the battery negative on the battery side of the sense resistor near the filter connection for SRP. The SRP and SRN have a common mode range to their supply from REGOUT and the VSS rail. When moving away from the recommended level due to high current or a buffer amplifier, the OCD and SCD may still trip, but accuracy could be compromised.

9.1.6 TSn Pins

The TSn pins must connect with a thermistor or resistor to the reference power pin for the associated cell group, as shown in the applications diagram. A resistor must be connected for normal operation even if external temperature measurement is not used. When thermistors are removable, they should be substituted with a test